Activity 6

Case Analysis – Avoiding Misleading Proportions

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Question 1: Truncated Y-Axis in Sales Growth

1. Why is the chart misleading?

The bar chart starts its Y-axis at **90** instead of **0**. Because of this, the difference between January (95 lakhs) and June (105 lakhs) appears much larger than it actually is.

The actual increase is:

- January sales = ₹95 lakhs
- June sales = ₹105 lakhs
- Increase = ₹10 lakhs

Percentage increase:

95105−95×100=10.53%

Therefore, the increase is only about **10.5%**, but the chart makes it look much larger.

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2. Visualization principle that has been violated

The principle of **honest and accurate representation of data** has been violated.

The following rules have not been followed:

- The axis scale should not distort the data.
- The visualization should accurately represent the true relationship between values.
- Bar charts should usually begin from zero to avoid exaggeration.

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3. Recommended visualization method

The company can use one of the following methods:

- A **bar chart** with the Y-axis starting at **0**.
- A **line chart** to show monthly sales trends.
- Clear labels and percentage values can also be added.

A line chart would be the best choice because it clearly shows gradual growth over time.

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4. Influence on business decisions

A misleading chart can affect decision-making in several ways:

- Managers may believe that sales have increased dramatically.
- The company might increase production unnecessarily.
- The marketing team may receive excessive funding.
- Investors may form incorrect conclusions about company performance.
- Resources may be allocated inefficiently.

Program

```
# Question 1: Truncated Y-Axis in Sales Growth


# Data

month <- c("January", "February", "March",

       "April", "May", "June")


sales <- c(95, 97, 99, 101, 103, 105)


# ------------------------------

# Misleading Bar Chart

# ------------------------------


barplot(sales,

     names.arg = month,

     ylim = c(90, 110),

     main = "Misleading Sales Chart",

     xlab = "Month",

     ylab = "Sales (₹ Lakhs)")


# ------------------------------

# Correct Bar Chart

# ------------------------------
```

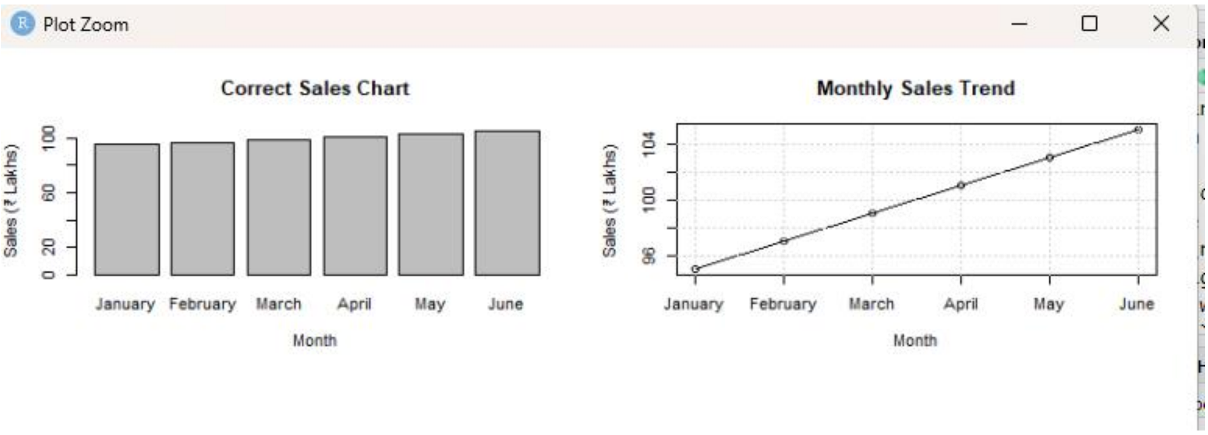
```r
barplot(sales,
        names.arg = month,
        ylim = c(0, 110),
        main = "Correct Sales Chart",
        xlab = "Month",
        ylab = "Sales (₹ Lakhs)")

# -------------------------------
# Recommended Line Chart
# -------------------------------

plot(sales,
     type = "o",
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales (₹ Lakhs)",
     main = "Monthly Sales Trend")

axis(1, at = 1:6, labels = month)
grid()
```

Output



Question 2: Pie Chart with Incorrect Proportions

1. Correct percentage calculation

Total number of students

420+280+200+100=1000

Percentage calculation

Percentage=(Total Students Department Students)×100

Department	Students	Percentage
Computer Science	420	42%
Electronics	280	28%
Mechanical	200	20%
Civil	100	10%

Correct proportions

- Computer Science – **42%**
- Electronics – **28%**
- Mechanical – **20%**
- Civil – **10%**

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2. Why is the chart misleading

The chart shows the following proportions:

- Computer Science – **50%**

- Electronics – **30%**
- Mechanical – **15%**
- Civil – **15%**

These percentages do not represent the actual data.

As a result:

- Computer Science appears larger than its actual value.
- Civil Engineering appears much larger than it should be.
- Mechanical Engineering appears smaller than its actual value.
- The relationship between the departments is incorrectly represented.

This violates the principle of **accuracy in data visualization**.

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3. Alternative visualization

A **bar chart** is the best alternative because it:

- Provides easier comparisons.
- Displays exact values.
- Reduces the possibility of visual distortion.
- Clearly shows differences between departments.

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4. Effects on institutional planning

An incorrect chart may lead to several problems:

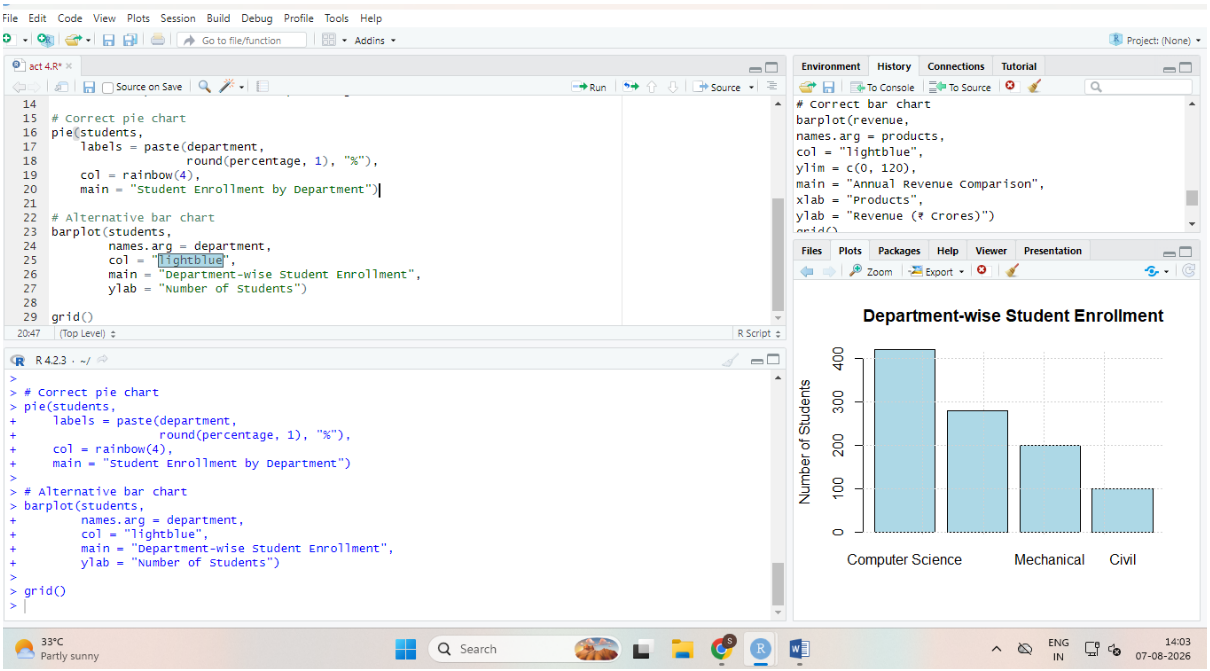
- Additional funding may be allocated to the wrong department.
- Faculty recruitment decisions may be inaccurate.
- Classroom allocation may become inefficient.
- Laboratory resources may not be distributed properly.
- Future admission plans may be affected.

Program

```
# Data

department <- c("Computer Science",

          "Electronics",

          "Mechanical",

          "Civil")


students <- c(420, 280, 200, 100)
```

```
# Calculate percentages
percentage <- students / sum(students) * 100

# Display percentages
data.frame(department, students, percentage)

# Correct pie chart
pie(students,
   labels = paste(department,
               round(percentage, 1), "%"),
   col = rainbow(4),
   main = "Student Enrollment by Department")

# Alternative bar chart
barplot(students,
       names.arg = department,
       col = "lightblue",
       main = "Department-wise Student Enrollment",
       ylab = "Number of Students")

grid()
```

Output



Question 3: Unequal Bar Widths in Revenue Comparison

1. Graphical error

The graphical error is the use of **unequal bar widths** in the bar chart.

Normally, all bars in a bar chart must have the **same width** so that only the height represents the value.

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2. How unequal bar widths distort perception

Human eyes perceive both the **height** and **area** of bars.

Because Product D has a greater width, its area becomes much larger, causing viewers to believe that its revenue is much higher than that of Product A.

Actual revenue comparison

Product	Revenue (₹ Crores)
A	60
B	75
C	90

Product	Revenue (₹ Crores)
D	105

Revenue increase from A to D:

105−60=45

Percentage increase:

6045×100=75%

However, the chart may create the impression that Product D generates almost three times the revenue of Product A.

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3. Correct design principles

The following principles should be followed:

- Use bars with equal widths.
- Begin the Y-axis at zero.
- Maintain equal spacing between bars.
- Use accurate labels and titles.
- Avoid visual distortions.

A simple bar chart with equal-width bars is the best solution.

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4. Possible consequences

Misleading visualizations may lead to the following problems:

- Investors may overestimate Product D's performance.
- Incorrect investment decisions may be made.
- Company resources may be distributed improperly.
- Future revenue predictions may become inaccurate.
- Stakeholders may lose confidence in the company.

Program

```
# Revenue data

products <- c("A", "B", "C", "D")

revenue <- c(60, 75, 90, 105)
```

Correct bar chart

barplot(revenue,

names.arg = products,

col = "lightblue",

ylim = c(0, 120),

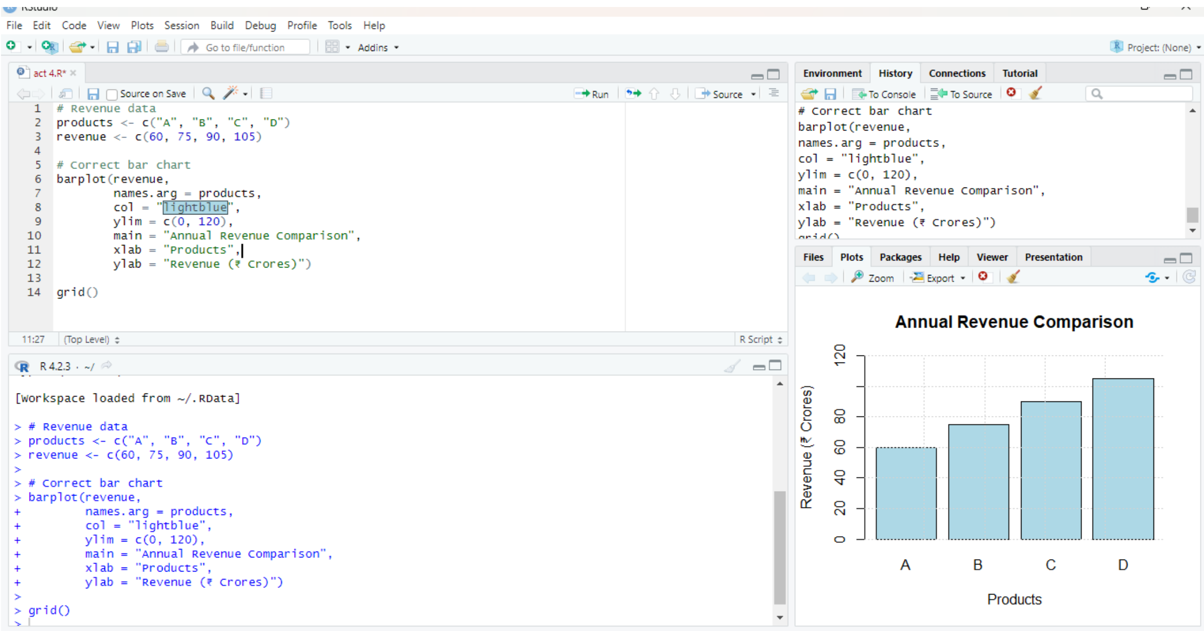
main = "Annual Revenue Comparison",

xlab = "Products",

ylab = "Revenue (₹ Crores)")

grid()

Output



Question 4: 3D Column Chart Distortion

1. Why are 3D charts misleading?

Three-dimensional charts often distort data because viewers perceive:

- Height

- Width
- Depth
- Perspective

As a result, bars located closer to the viewer appear larger, while bars farther away appear smaller.

In this case, the Hydro column appears much larger because of the viewing angle, even though the actual increase is relatively small.

This violates the principle of **accurate data representation**.

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2. Actual difference versus perceived difference

Actual values

- Wind = 500 GWh
- Hydro = 550 GWh

Actual increase

550−500=50

Percentage increase

50050×100=10%

Perceived increase

Because of the three-dimensional effect, viewers may interpret the increase as approximately **40%**.

Measurement	Value
Actual increase	10%
Perceived increase	40%

Therefore, the visual representation exaggerates the difference.

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3. Recommended visualization

The following alternatives can be used:

- Two-dimensional (2D) bar chart

- Horizontal bar chart
- Line graph
- Dot plot

A **2D bar chart** is the most suitable option because it accurately represents the values without introducing perspective distortion.

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4. Importance of avoiding visual distortion

Avoiding visual distortion is essential because:

- Investors rely on accurate information.
- Government agencies use the data for policy decisions.
- Scientists use the information for research purposes.
- Incorrect information can lead to poor resource allocation.
- Public trust in sustainability reports may decrease.

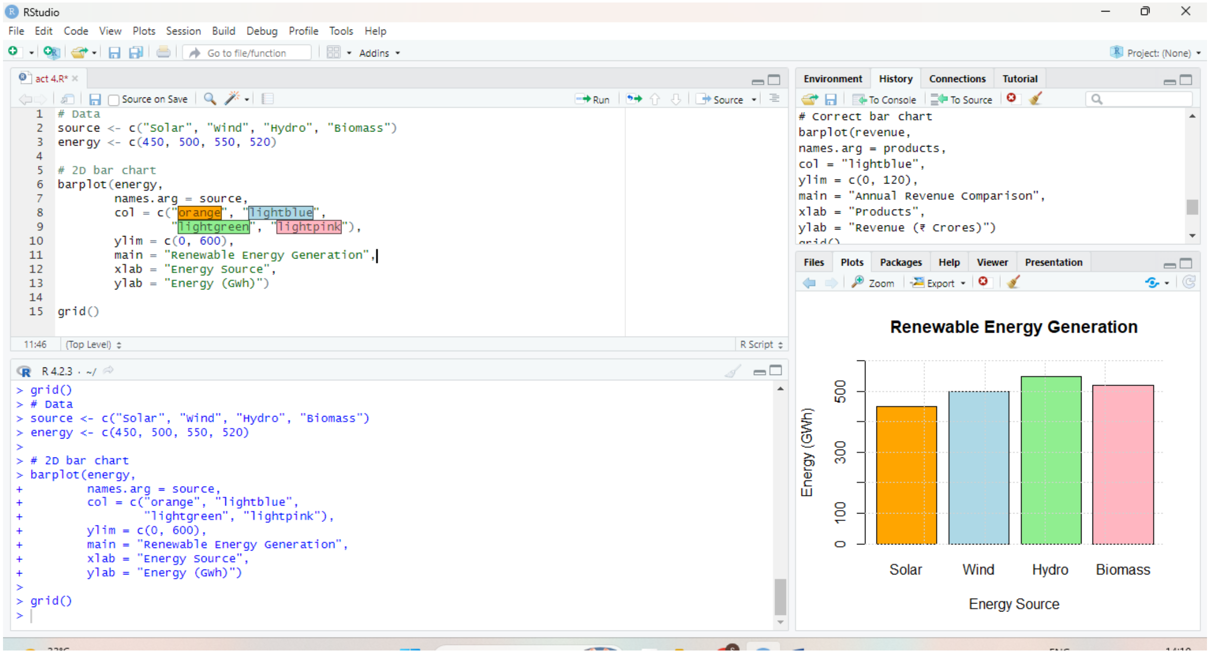
Accurate visualizations help organizations maintain transparency and credibility.

Program

```
# Data

source <- c("Solar", "Wind", "Hydro", "Biomass")

energy <- c(450, 500, 550, 520)



# 2D bar chart

barplot(energy,

    names.arg = source,

    col = c("orange", "lightblue",

        "lightgreen", "lightpink"),

    ylim = c(0, 600),

    main = "Renewable Energy Generation",

    xlab = "Energy Source",

    ylab = "Energy (GWh)")
```

grid()

Output



Question 5: Bubble Chart with Incorrect Scaling

1. Why is scaling the radius misleading?

In a bubble chart, people compare the **area** of the circles rather than the radius.

If the radius is directly proportional to the number of cases, the area increases much more rapidly because

Area=πr2

This makes the larger bubbles appear much bigger than they should.

For example:

- District A = 150 cases
- District E = 550 cases

Actual ratio:

150550=3.67

However, if the radius is scaled directly, the area ratio becomes

(150550)2=13.44

Thus, District E appears almost **13 times larger** than District A, even though the actual difference is much smaller.

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2. Correct method for proportional bubble sizing

The radius should be proportional to the square root of the data value.

r∝Cases

This method ensures that the area of the bubble accurately represents the number of cases.

The following steps should be followed:

- Calculate the square root of each value.
- Use these values as the radii of the circles.
- Keep the scale consistent for all districts.

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3. Alternative visualization methods

The following charts can be used:

- Bar chart
- Pie chart
- Line graph
- Dot plot

A **bar chart** is the most suitable option because it allows accurate comparison of case numbers.

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4. Impact on public health resource allocation

A misleading chart can create several problems:

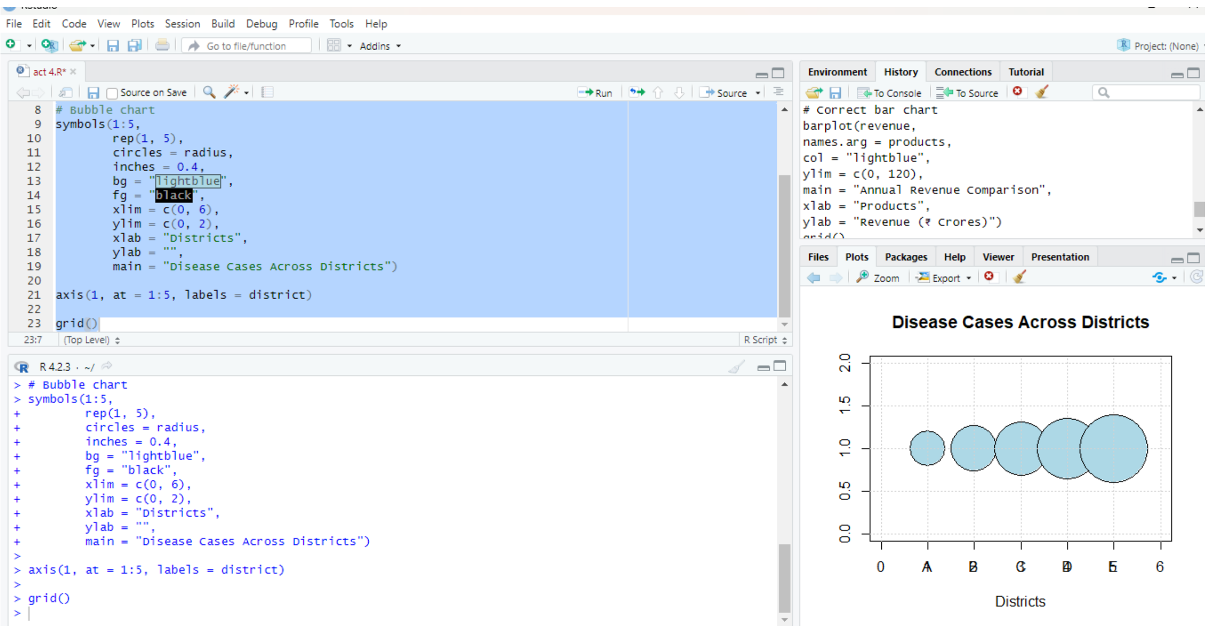
- Health officials may overestimate the severity of disease outbreaks.
- More medicines and equipment may be allocated to a particular district.
- Funds may be distributed unevenly.
- Other districts may receive fewer resources.
- Incorrect decisions may affect public health planning.

Program

Data

```
district <- c("A", "B", "C", "D", "E")

cases <- c(150, 250, 350, 450, 550)


# Correct bubble size calculation

radius <- sqrt(cases)


# Bubble chart

symbols(1:5,

        rep(1, 5),

        circles = radius,

        inches = 0.4,

        bg = "lightblue",

        fg = "black",

        xlim = c(0, 6),

        ylim = c(0, 2),

        xlab = "Districts",

        ylab = "",

        main = "Disease Cases Across Districts")


axis(1, at = 1:5, labels = district)


grid()
```

output


File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
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Source on Save
Run
Source
Bubble chart
symbols(1:5,
rep(1, 5),
circles = radius,
inches = 0.4,
bg = "lightblue",
fg = "black",
xlim = c(0, 6),
ylim = c(0, 2),
xlab = "Districts",
ylab = "",
main = "Disease Cases Across Districts")
axis(1, at = 1:5, labels = district)
grid()
23:7
(Top Level)
R Script
R 4.2.3 · ~/
> # Bubble chart
> symbols(1:5,
+ rep(1, 5),
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+ inches = 0.4,
+ bg = "lightblue",
+ fg = "black",
+ xlim = c(0, 6),
+ ylim = c(0, 2),
+ xlab = "Districts",
+ ylab = "",
+ main = "Disease Cases Across Districts")
>
> axis(1, at = 1:5, labels = district)
>
> grid()
Environment History Connections Tutorial
To Console
To Source
Correct bar chart
barplot(revenue,
names.arg = products,
col = "lightblue",
ylim = c(0, 120),
main = "Annual Revenue Comparison",
xlab = "Products",
ylab = "Revenue (₹ Crores)")
Files Plots Packages Help Viewer Presentation
Zoom
Export
Disease Cases Across Districts
0.0
0.5
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Districts